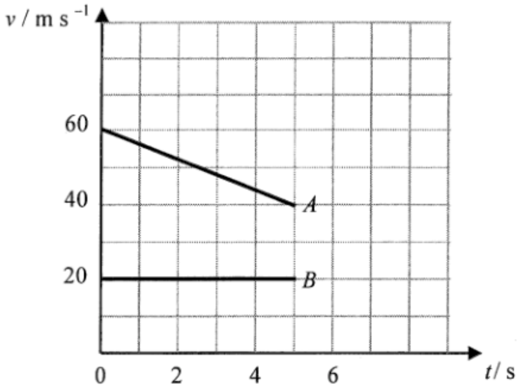
4. (a) (i) $v = u + at$

$= 60 + (-4)5$ 1M

$= 40 \text{ m s}^{-1}$ 1A 2

(ii)



1A 1

(iii) $s_A = \left(\dfrac{60+40}{2}\right)(5) = 250 \text{ m}$ 1M

$s_B = (20)(5) = 100 \text{ m}$

$x = 250 - 100$ 1M

$= 150 \text{ (m)}$ 1A 3

[equals to the area between the two graphs]

(b) (i) $m\,u_A + m\,u_B = (m+m)V$ 1M

$40 + 20 = 2V$

$V = 30 \text{ m s}^{-1}$ 1A 2

(ii) $F = \dfrac{mV - mu_A}{\Delta t}$

$= (5000)\dfrac{(30-40)}{0.2}$ 1M

$= -250000 \text{ N}$ 1A

Impact force is opposite to the direction of travel of *A* (to the left / backwards / negative) 1A 3

(iii) The thermometer would be in direct contact with the cooler air and would cool down quickly. 1A
The temperature reading would be less than the actual soil temperature. 1A

2

2. Measure the mass of a bullet m and the mass of the trolley with plasticine M. 1A
Fire the bullet towards the plasticine. 1A
Read the speed of the trolley v immediately after the bullet hit the plasticine. 1A

The speed of the bullet u is given by $u = \frac{M+m}{m}v$. 1A

Precaution: (Any **ONE**)
- The bullet should be fired close to the plasticine. 1A
- The bullet should be fired along the direction of travel of the trolley. 1A
- The track must be horizontal / friction compensated. 1A

5

(a) $\frac{(c_{rms})_f}{c_{rms}} = \sqrt{\frac{T_f}{T}}$ 1M

$\frac{c_{\text{r.m.s.}} \text{ at } 350\text{K}}{c_{\text{r.m.s.}} \text{ at } 300\text{K}} = \sqrt{\frac{350}{300}}$

$= 1.08$ 1A

2

(b) The speed of the gas molecules increases. They collide more frequently and violently with the wall of the container. 1A+1A
Thus, the pressure increases.

2

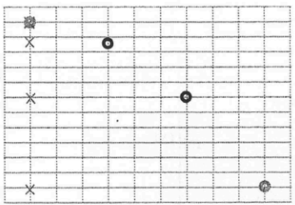
4. (a) (i) By $s = ut + \frac{1}{2}gt^2$

$0.11 = \frac{1}{2}g(0.05 \times 3)^2$ 1M

$g = 9.78 \text{ m s}^{-2}$ 1A

2

(ii) (1)



1A Correct horizontal positions
1A Correct vertical positions

2

(2) $v_x = 1 \text{ m s}^{-1}$

$v_y = u_y + gt$

$= 0 + 9.78 \times (0.05 \times 3)$ 1M

$= 1.47 \text{ m s}^{-1}$

$v = \sqrt{v_x^2 + v_y^2}$ 1M

$= \sqrt{1^2 + 1.47^2}$

$= 1.78 \text{ m s}^{-1}$ 1A

3

(b) The air resistance acting on the ball increases as its speed increases. 1A
When the air resistance equals to the weight of the ball, 1A

net force acting on the ball becomes zero, by Newton's first law of motion, the ball travels with constant speed. 1A

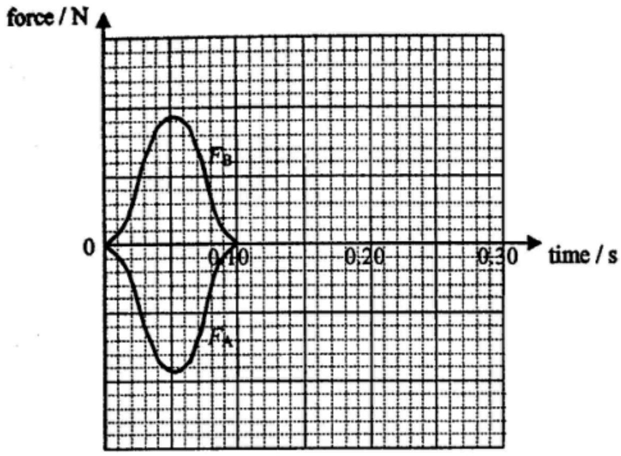
Or
net force acting on the ball becomes zero, by Newton's second law of motion, the ball will not accelerate further and travels with constant speed. 1A

3

2018 Q4

		Marks	Remarks
	$0 = 25^2 + 2(-9.81\sin 30^\circ)D$ $D \approx 63.7$ m	1A	(or 31.25 m for $g = 10$ m s^{-2}) Accept D = 62.0 m to 64.0 m
		2	
4. (a) (i)	$\text{K.E.} + \text{P.E.} = \frac{1}{2}(0.3)(4)^2 + (0.3)(9.81)(0.2)$	1M+1M	
	$= 2.4 + 0.5586 = 2.9886$ J ≈ 2.99 J (3.0 J for $g = 10$ m s^{-2})	1A	$= 2.4 + 0.6 = 3.0$ J for $g = 10$ m s^{-2}
		3	
(ii)	As the spring gun is fixed, there is external force acting on the system / the gun,	1M	
	total momentum (of the spring gun and cannon ball) is not conserved.	1A	
		2	
(b)	Vertical : $s = ut + \frac{1}{2}at^2$	1M	Or $\frac{t_f}{2} = \frac{u\sin\theta}{g} = \frac{4\sin 50^\circ}{9.81}$
	$0 = (4\sin 50^\circ)\,t_f - \frac{1}{2}(9.81)\,t_f^2$ $t_f = 0.624705$ s (0.612836 s for $g = 10$ m s^{-2}) ≈ 0.625 s (0.613 s for $g = 10$ m s^{-2})	1A	Accept t_f = 0.61 s to 0.63 s
	Horizontal : $R = 4\cos 50^\circ \times t_f = 4\cos 50^\circ \times 0.625$	1M	Or $R = u\cos\theta \times \frac{2(u\sin\theta)}{g}$
	$= 1.606210$ m ≈ 1.61 m (1.57 m for $g = 10$ m s^{-2})	1A	Accept R = 1.56 m to 1.62 m
		4	
(c)	t_f increases	1A	
	since the initial vertical velocity / component is greater.	1A	
		2	

10. (a) 1. Use the protractor to measure or mark a specific angle at which the ball is released. Make sure the angle is measurable and recorded.	1A	
2. Hold the ball at the marked angle and release it from rest with the string taut, allowing it to swing like a pendulum.	1A	
3. Measure the angle at which the ball reaches its highest point and record the value. Or Observe the angle at which the ball swings back to the original releasing position.	1A	
4. Compare the measurements of the ball's initial angle of release to the angle it reaches on the opposite side of the swing. (The angles equal implies total mechanical energy is conserved.)	1A	
	4	
(b) (i) (I) The tension forces are vertical / perpendicular to the direction of collision, the condition of no external forces along the colliding / horizontal direction is still fulfilled, the law can be applied. Or The tensions in the string acting on the spheres are balanced by the weights of the spheres so that the net force acting on the system is zero.	1A	
	1	
(II) By conservation of total momentum, v_E is $2 \times 0.50 = 1.0\ \text{m s}^{-1}$	1A	
However, the final total kinetic energy would then be $\frac{1}{2}(0.020)(1.0)^2 = 0.01\ \text{J} > \frac{1}{2}(2 \times 0.020)(0.50)^2 = 5 \times 10^{-3} = 0.005\ \text{J}$	1M	
which is greater than the total kinetic energy before collision, thus it is not possible.	1A	
	3	
(ii) Not perfectly elastic as some kinetic energy is lost as sound / thermal energy, the total kinetic energy is not conserved.	1A+1A	

towards the right taken as positive), give trolley *A* a push and let it collide head-on with trolley *B*.	1A		
Force-time graphs: 	2A		
The force acting on (the less massive) trolley *B*, i.e. F_B, is always equal (in magnitude and opposite in direction) to that acting on trolley *A*, i.e. F_A, as they are an action and reaction pair / according to Newton's 3rd law.	1A		
		4	